



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2 -
Academic Year	I-III Semester (2025 26)
Faculty Name	Swati
Student Name	Kandula Poojitha Rani
Roll Number	2520030460
Branch	CSE
Course Outcome	

Case Study Topic: Retail Store Network Analysis using BFS and Kruskal Algorithm for RetailMind – Smart Retail Inventory & Sales Analytics System.

Work Done Summary: This case study implements BFS and Kruskal Algorithm in Java for retail store network analysis in the RetailMind – Smart Retail Inventory & Sales Analytics System. The program represents stores and supply routes using a graph structure. BFS is used to analyze store connectivity and network reachability, while Kruskal Algorithm is applied to build a minimum-cost distribution network using Minimum Spanning Tree (MST). Java classes and methods were developed for graph creation, traversal, edge sorting, and union-find operations. This approach improves routing efficiency, reduces transportation cost, and supports effective supply chain management in retail operations.

Implementation Details:

1. Created a **graph structure** to represent retail stores and supply routes.
2. Implemented **BFS traversal** to check store connectivity and network reachability.
3. Applied **Kruskal Algorithm** to find the **Minimum Spanning Tree (MST)**.
4. Used **sorting and union-find operations** to avoid cycles and build the minimum-cost network.
5. Integrated the model with **RetailMind** to optimize store connections and distribution routes.
6. Improved network analysis and routing efficiency using graph algorithms.

```
BFS Traversal: 0 1 2 3
```

```
Minimum Cost Network:
```

```
2 - 3 : 4
```

```
0 - 3 : 5
```

```
0 - 1 : 10
```

```
Total Cost: 19
```

Result: Screenshots/Output:

```
=== Code Execution Successful ===
```

The BFS and Kruskal based network analysis system successfully models retail store connectivity in the RetailMind application. BFS helps analyze store connections and reachability, while Kruskal Algorithm identifies the minimum-cost distribution network. This improves routing efficiency, reduces operational cost, and supports better decision-making in retail supply management.

GitHub Link:

<https://github.com/kandulapoojitharani-ai/Retail-Store-Network-Analysis-using-BFS-and-Kruskal-Algorithm-for-RetailMind>

Student Signature	Faculty Signature